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Research Article

Multibridge Inference Structural Health Monitoring (MISHM): A Drive-By Crowdsensing Approach at the Network Level

Jiangyu Zeng, Qipei Mei, and Mustafa Gül

Department of Civil & Environmental Engineering, University of Alberta, Edmonton, Alberta, Canada

Correspondence should be addressed to Mustafa Gül; mustafa.gul@ualberta.ca

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As aging bridge infrastructure poses increasing safety risks, there is a critical need for reliable and scalable Structural Health Monitoring (SHM) systems. Traditional SHM methods, which rely on fixed sensor networks and assessments of individual bridges, face significant challenges in scalability, cost, and efficiency—particularly in complex urban environments. To address these limitations, this study introduces the Multibridge Inference SHM (MISHM) framework. MISHM leverages drive-by monitoring and crowdsensing to observe multiple bridges simultaneously. It employs a feature-based analysis using Mel-frequency cepstral coefficients (MFCCs) and Kullback–Leibler (KL) Divergence to identify structural changes. Here, "inference" refers to drawing conclusions about the health of each individual bridge by comparing patterns and features gleaned from the entire network, rather than relying on isolated measurements. By making multiple comparisons across all monitored structures, MISHM enhances fault tolerance, reduces missed detections, and offers a scalable solution for smart city infrastructure monitoring. This framework represents a vital advancement in SHM systems, addressing the evolving needs of large-scale urban infrastructure management.

1. Introduction

Structural Health Monitoring (SHM) is critical for maintaining the safety and longevity of infrastructure, particularly bridges [1], which serve as essential components of modern urban networks. Many bridges worldwide are approaching or exceeding their intended service life, necessitating reliable monitoring systems to prevent failures and maintain serviceability. For instance, in the United States, 42% of over 617,000 bridges are at least 50 years old, and 7.5% are considered structurally deficient [2], highlighting the urgent need for effective SHM solutions [3–6]. In the context of rapidly urbanizing smart cities, the demand for scalable and cost-efficient infrastructure monitoring solutions is becoming increasingly critical to support urban growth and ensure public safety [7].

Traditional SHM methods often rely on fixed sensor networks installed directly on bridge structures [8–12], which can be costly and logistically challenging, especially for bridges in remote or inaccessible locations. The installation and maintenance of these sensors require direct access to the bridge, leading to increased expenses and potential disruptions to normal operations. Additionally, because traditional SHM methods typically concentrate on monitoring a single bridge at a time, deploying fixed sensors becomes labor-intensive and often unfeasible when attempting large-scale monitoring of extensive bridge networks [13]. These challenges are further magnified in rapidly urbanizing smart cities, which focus heavily on the growing need for scalable and cost-efficient bridge monitoring.

In recent years, there has been growing interest in alternative SHM techniques that are more cost-effective, less disruptive to normal operations, and more adaptable to large-scale deployments. Among these innovations are drive-by monitoring methods [14–21], which leverage data collected from vehicles as they traverse bridges. Yang et al. [22] pioneered the concept of extracting bridge frequencies from the dynamic response of a passing vehicle, laying the

groundwork for what would become a significant area of research. This method involves using the vehicle as a sensor to infer the health of the bridge, thereby eliminating the need for stationary sensors on the bridge itself. Building upon this, Shirzad-Ghaleroudkhani and Gül [23] employed inverse filtering techniques to isolate bridge frequencies from vehicle data, enhancing structural characteristic identification with minimal on-site equipment. Subsequent studies have further explored the potential and practicality of drive-by methods. McGetrick et al. [14] introduced a novel algorithm for identifying bridge stiffness solely from vehicle acceleration data, demonstrating its feasibility through a laboratory experiment. In a related effort, McGetrick et al. [24] investigated the use of readily available smartphone technology and GNSS to implement drive-by monitoring in the field, revealing that smartphones can serve as a cost-effective alternative to traditional accelerometers and GPS systems. More recently, Sadeghi Eshkevari et al. [25] presented two methodologies to achieve comprehensive bridge modal identification using data exclusively collected by moving vehicles. By effectively disentangling the bridge's vibration signature from vehicle dynamics and road surface irregularities, these approaches underscore the growing sophistication of drive-by techniques and their increasing alignment with real-world SHM applications.

More recently, SHM systems have benefited from the integration of crowdsensing [26–31], further improving their efficiency and scalability. Crowdsensing, which leverages the widespread availability of smartphones and vehicles as mobile sensors, bypasses the need for expensive, dedicated sensor networks by utilizing data collected from numerous passing vehicles. This approach offers substantial advantages in terms of cost-effectiveness, scalability, and spatial coverage, making it particularly suitable for monitoring large populations of bridges. In this context, Mei and Gül [32] proposed a framework that fuses crowdsensed smartphone data with advanced analytics, such as Mel-frequency cepstral coefficients (MFCC) and Kullback–Leibler (KL) Divergence, to monitor individual bridges, while Shirzad-Ghaleroudkhani and Gül [33] refined these techniques using inverse filtering to better address operational effects. Complementary efforts have employed alternative time-frequency analysis methods. For instance, Matarazzo et al. [34] applied the synchrosqueezed wavelet transform to asynchronously aggregated smartphone data, estimating the bridge's most probable modal frequencies under real-world conditions, and Eshkevari et al. [35] introduced a continuous-wavelet-based method to identify both natural frequencies and absolute mode shapes, including torsional modes, from mobile sensor data. Collectively, these studies demonstrate the growing sophistication of crowdsensing-assisted SHM, as well as the potential for scalable, data-driven monitoring solutions that accommodate complex operational environments.

Extending these advancements, this research introduces the Multibridge Inference SHM (MISHM) framework, a novel approach that leverages information from multiple bridges within a network (Figure 1). Rather than examining each structure in isolation, MISHM correlates the dynamic responses of all bridges, enabling a collective perspective on infrastructure health. Theoretically, this network-level inference reduces the likelihood of overlooked damage, as any local indications are evaluated against multiple points of reference. As the size of the monitored network grows, so does the robustness of the analysis, making MISHM particularly well-suited for large-scale applications in smart city environments [36, 37].

MISHM integrates a feature-based analysis methodology, utilizing MFCC features combined with KL Divergence to evaluate the dynamic responses of bridges collected from vehicles passing over them. This method provides unique insights into the health of multiple bridges in a network, contributing to a robust SHM framework aligned with the needs of smart city infrastructure management.

The primary objective of this research is to develop a scalable SHM framework capable of improving detection accuracy and effectively monitoring large-scale networks of bridges. The MISHM approach addresses the critical need for scalable monitoring solutions in urban environments, particularly within the framework of smart cities. By leveraging redundancy through multiple comparisons across the network, MISHM aims to reduce missed detections (false negatives), thus providing a robust monitoring solution. The methodology has been validated through comprehensive simulations, establishing a foundation for future implementation in actual bridge networks.

By integrating advanced data analytics with crowdsensing techniques, this research specifically demonstrates the effectiveness of the MISHM framework for monitoring multiple bridges simultaneously within urban environments. The proposed approach utilizes MFCCs and KL Divergence to analyze dynamic responses, enhancing the accuracy and scalability of SHM. By reducing the dependency on individual baselines and incorporating multibridge inference, MISHM offers a robust solution for identifying structural anomalies in densely populated areas. The findings presented in this paper provide a foundational basis for future real-world implementation and are expected to support the adoption of multibridge health monitoring, which is particularly suited for smart cities where scalability and efficiency are crucial.

The paper is organized as follows: The MISHM approach, incorporating MFCCs and KL Divergence for feature-based Relative Damage Index (RDI) analysis, is described in Section 2. The computational study, including the Vehicle-Bridge Interaction (VBI) model and data preprocessing, is presented in Section 3. The effectiveness of the proposed method in detecting damage is evaluated and discussed in Section 4, and key findings along with future research directions are summarized in Section 5.

2. Methodolody

The novelty of this drive-by bridge health monitoring framework lies in its ability to simultaneously consider multiple bridges within a defined area, employing cross-comparisons among them to enable large-scale, network-level assessments that align with the demands of smart city infrastructure management. The overall flowchart of the method is elaborated in Figure 2. The MISHM framework

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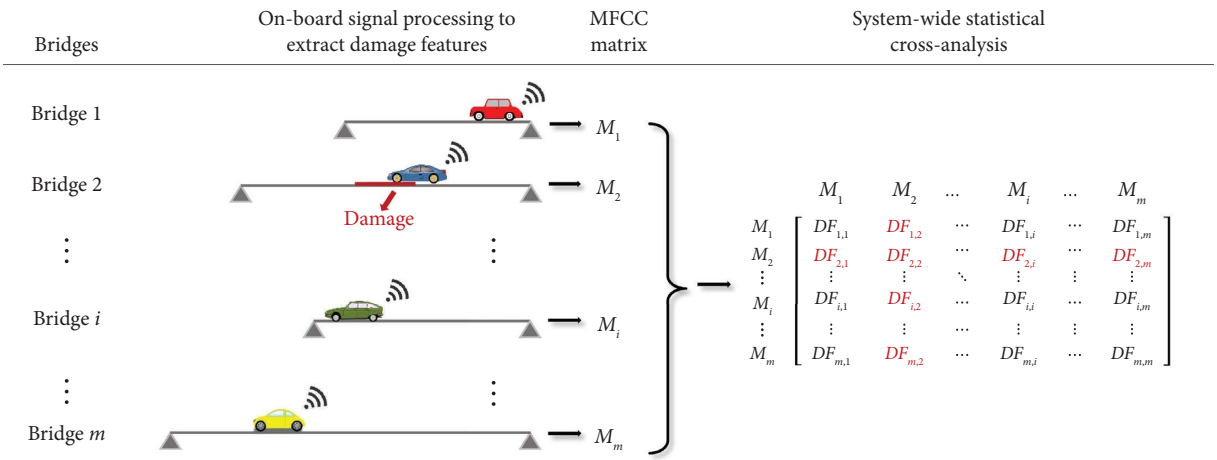


FIGURE 1: Overview of the MISHM framework for multiple-bridge monitoring (cliparts from [38]).

uses vibration signals collected across baseline, validation, and testing states to extract MFCCs (discussed in detail in Section 2.1), which capture key damage features of the dynamic responses of bridges. These features are then used to compute Divergence Factors (described in Section 2.2) through KL Divergence, enabling cross-comparisons between bridges to detect anomalies. Finally, an RDI (defined in Section 2.3) is calculated for each bridge, identifying damage by comparing this RDI to a threshold established during the validation phase.

2.1. MFCCs. MFCCs are inspired by the human auditory system's frequency response, which is inherently more sensitive to changes at lower frequencies than higher ones [39]. This non-linear sensitivity is effectively captured by the Mel scale, originally developed to approximate human sound perception. In bridge health monitoring applications, raw vibration signals collected either by smartphones securely mounted in moving vehicles or by vehicle-integrated sensors undergo several processing steps to generate MFCCs. This procedure highlights frequency-domain features particularly sensitive to the structural characteristics and integrity of bridges [40].

Recent experimental studies have validated the effectiveness of MFCC-based approaches in detecting subtle structural anomalies in bridges. Mei et al. [41] demonstrated through laboratory experiments that MFCCs effectively capture dynamic characteristics sensitive to changes in bridge conditions. Shirzad-Ghaleroudkhani and Gül [33] further developed and improved upon Mei et al.'s methodology by applying inverse filters to raw signals and validating the method's performance through both laboratory and real-life field experiments, successfully capturing distinct patterns in the frequency spectrum related to structural integrity. Similarly, Li et al. [42] successfully applied MFCC-based drive-by damage detection methods validated through practical field experiments, further confirming MFCCs' practical applicability and sensitivity. Additionally, de Souza et al. [43] explored the feasibility of MFCCs for damage detection in high-speed railway bridges under realistic conditions, providing further support for the robustness and reliability of MFCCs in practical SHM scenarios. Collectively, these experimental validations underscore the effectiveness and suitability of MFCCs as a powerful tool for SHM.

The overall procedure for MFCC analysis used in this study is illustrated in Figure 3, with detailed computational steps described as follows:

1. To reduce noise, a Gaussian low-pass filter with a sigma of 0.5 was applied to the original acceleration data $x(t)$. This filtering process, denoted as $\tilde{x}(t)$, suppressed high-frequency noise while preserving essential signal characteristics. These preprocessed signals were then recorded and set as the baseline state.
2. The filtered signal $\tilde{x}(t)$ is subjected to a Fourier transform to convert it from the time domain to the frequency domain. This transformation yields the frequency spectrum of the signal, which is resampled to focus on the frequency range from 0 to 20 Hz with a desired number of points. This range is chosen to optimize computational efficiency, as the fundamental frequencies of bridges are typically below 20 Hz [23]. The resampled signal is denoted as $\tilde{x}_f(f)$, where f represents the frequency domain.
3. A Mel filter bank is then established to process the frequency domain signal. The Mel scale, which approximates the human ear's response to different frequencies, is used to convert frequency values into a perceptually relevant scale. The Mel filter bank is constructed through the following procedures [40]:
   i. To evenly space Mel frequencies over a specified range, the range limits are first converted from Hertz to the Mel scale using:

$$m_{\text{first}} = 5 \cdot \ln\left(1 + \frac{f_{\text{first}}}{5}\right), \tag{1}$$

$$m_{\text{last}} = 5 \cdot \ln\left(1 + \frac{f_{\text{last}}}{5}\right), \tag{2}$$

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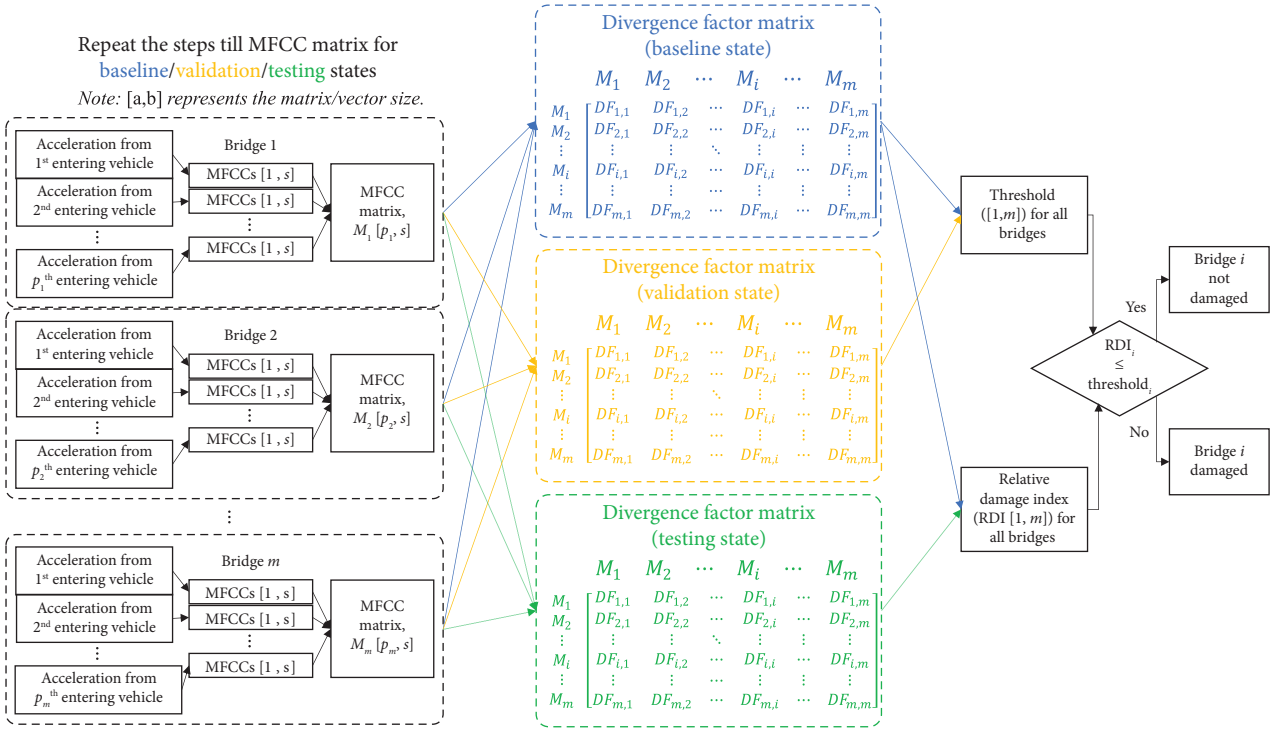


FIGURE 2: Flowchart of the proposed MISHM framework.

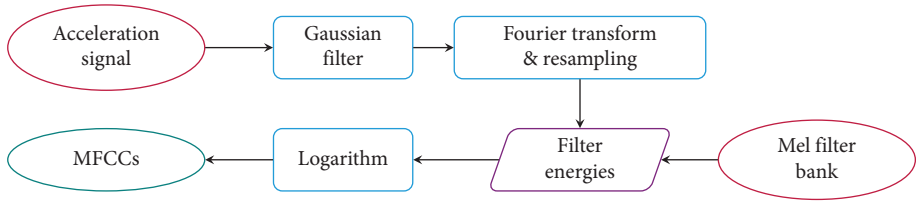


FIGURE 3: Flowchart of the MFCC feature extraction process.

where m_{first} and m_{last} are the Mel frequencies corresponding to the first and last Hertz frequencies f_{first} and f_{last}, respectively.

ii. The Mel increment Δm is computed to evenly space the Mel frequencies:

$$\Delta m = \frac{m_{\text{last}} - m_{\text{first}}}{N - 1}, \tag{3}$$

where N denotes the number of key frequencies. This increment determines the step size in the Mel scale, ensuring that the key frequencies are evenly distributed across the Mel scale, which corresponds to a nonuniform distribution in Hertz.

iii. Using the computed Mel increment, the key Mel frequencies are calculated by progressively adding the increment to the starting Mel frequency:

$$m_i = m_{\text{first}} + (i - 1) \cdot \Delta m, \tag{4}$$

where m_i represents the i-th key Mel frequency, and i ranges from 1 to N.

iv. Once the key Mel frequencies are obtained, they are converted back to Hertz to determine the actual frequencies used in the filter bank:

$$f_i = 5\left(e^{m_i/5} - 1\right). \tag{5}$$

This conversion uses the inverse of the Mel scale formula to translate the evenly spaced Mel frequencies into Hertz values, as the filter bank operates in the Hertz domain.

v. To construct the Mel filter bank, triangular filters are created centered around each key frequency. The filter values at each bin k are determined by:

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$$H_i(k) = \begin{cases} \dfrac{2 \cdot (k - f_{\text{left}})}{(f_{\text{center}} - f_{\text{left}}) \cdot (f_{\text{right}} - f_{\text{left}})}, & \text{for } f_{\text{left}} \le k < f_{\text{center}}, \\ \dfrac{2 \cdot (f_{\text{right}} - k)}{(f_{\text{right}} - f_{\text{left}}) \cdot (f_{\text{right}} - f_{\text{center}})}, & \text{for } f_{\text{center}} \le k < f_{\text{right}}, \\ 0, & \text{otherwise}, \end{cases} \tag{6}$$

where $H_i(k)$ denotes the value of the triangular filter at bin k, f_{center} is the center frequency of the filter, and f_{left} and f_{right} are the left and right boundaries of the filter, respectively.

It is important to note that the triangular filters used in this study include a normalization step, which differs from the approach described in Mei and Gül [40]. This normalization step is consistent with that applied by Shirzad-Ghaleroudkhani and Gül [33], where it was noted that the amplitude of each triangular weighting function in the Mel filter bank decreases with increasing frequency. While the term "filter" is used here following common practice in Mel-frequency analysis, these filters do not completely remove or attenuate higher-frequency components as traditional filters might. Instead, they act as frequency weighting functions, assigning greater emphasis (higher amplitude) to lower-frequency components and relatively less emphasis to higher-frequency components. By ensuring that each filter's values sum to unity across all frequency bins, the normalization balances the energy distribution, effectively reducing the relative influence of higher-frequency components and enhancing sensitivity to subtle changes in lower frequencies, which are critical indicators of structural integrity in SHM. The differences between the normalized and non-normalized triangular weighting functions (Mel filters) are illustrated in Figure 4.

4. After the Mel filter bank $H_k(k)$ has been constructed and the resampled signal $\tilde{x}_f(f)$ has been obtained, the logarithms of the filter energies are calculated by applying the filter bank to the resampled signal. This involves multiplying the filter bank matrix with the resampled signal vector, effectively summing the contributions of the signal within each filter's range. This can be mathematically expressed as follows:

$$\log(E_i) = \log\left(\sum_{k=1} H_i(k) \cdot \tilde{x}_f(f_k)\right), \tag{7}$$

where E_i represents the i-th filter energy.

5. Finally, the Discrete Cosine Transform (DCT) is employed to convert the logarithm of the filter energies into MFCCs:

$$\text{MFCC}_j = \sum_{i=2}^{N-1} \log(E_i) \cdot \cos\left(\frac{\pi}{N-2}(i - 1.5)j\right), \tag{8}$$

where MFCC_j is the j-th MFCC, and j ranges from 2 to $N - 1$.

2.2. Divergence Factor. The key innovation of this paper lies in the cross-analysis approach, which facilitates network-level monitoring of multiple bridges simultaneously. Rather than evaluating each structure in isolation, the framework compares every bridge to every other bridge for both baseline and unknown states. This approach is particularly suited for large-scale bridge networks, enabling system-wide insights and interactions to be captured more effectively.

To quantify this cross-comparison, the concept of the Divergence Factor is introduced. The Divergence Factor measures the distribution of divergences between pairs of bridges. By organizing these Divergence Factors into a matrix, with dimensions corresponding to the number of bridges, the Divergence Factor Matrix is constructed. This matrix serves as the foundation for deriving the Damage Index for each bridge, as will be discussed in the following subsection.

The steps to derive the Divergence Factors are as follows:

1. Let M_i denote the MFCCs matrix for Bridge i, where $i = 1, 2, \ldots, m$. Each M_i has a size of $s \times p_i$, where s is the number of coefficients retained from the MFCCs, and it must be consistent across all matrices; p_i is the number of random vehicles that traverse Bridge i.
2. The KL Divergence between M_i and M_j is defined as follows:

$$D_{\text{KL}}\left(M_i \| M_j\right) = \frac{1}{2}\left[\text{tr}\left(\Sigma_j^{-1}\Sigma_i\right) + \left(\mu_j - \mu_i\right)^T \Sigma_j^{-1}\left(\mu_j - \mu_i\right) - s + \ln\left(\frac{\det \Sigma_j}{\det \Sigma_i}\right)\right], \tag{9}$$

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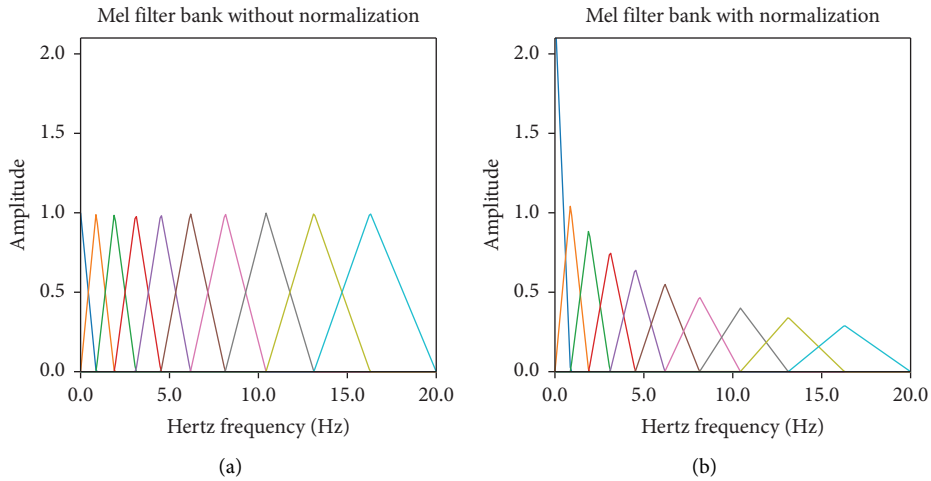


FIGURE 4: Comparison of Mel filter banks: normalised vs. non-normalised triangular filters. (a) Mel filter bank without normalisation. (b) Mel filter bank with normalisation.

where μ_i and Σ_i represent the mean vector and covariance matrix of M_i, respectively; μ_j and Σ_j denote the mean vector and covariance matrix of M_j, respectively; tr indicates the trace of a matrix; and det signifies the determinant of a matrix.

3. Each entry $DF[i, j]$ represents the Divergence Factor between M_i and M_j and is computed as follows:

$$DF[i, j] = \ln\left(D_{\mathrm{KL}}\left(M_i \middle\| M_j\right) + e\right) - 1. \tag{10}$$

2.3. RDI. In this subsection, a novel approach is introduced to quantify bridge damage within a network context by developing a RDI. Instead of assessing a single bridge independently, the RDI measures damage in relation to other bridges in the system. By placing each bridge's condition within a broader network perspective, this comparative metric highlights how each bridge's current state compares to the collective health of the network. As a result, the RDI can support more informed maintenance and repair planning, helping to identify which bridges warrant closer attention within a large-scale monitoring framework.

The procedures to define the RDI and establish corresponding thresholds are as follows:

1. Compute the Divergence Factor matrices for the baseline (intact) and unknown states. These matrices are denoted as $\mathrm{DF}_{\mathrm{baseline}}$ and $\mathrm{DF}_{\mathrm{unknown}}$, respectively.
2. Compute the Percentage Difference Matrix (PDM) $\Delta DF[i, j]$, which quantifies the change in Divergence Factors between the baseline and unknown states. To avoid numerical instability, especially when the baseline divergence factor values are close to zero, a stabilizing constant ε is introduced. This constant is determined as the median of the off-diagonal elements in the baseline divergence factor matrix. The stabilized percentage difference is computed as follows:

$$\Delta \mathrm{DF}[i, j] = \frac{\mathrm{DF}_{\mathrm{unknown}}[i, j] - \mathrm{DF}_{\mathrm{baseline}}[i, j]}{\mathrm{DF}_{\mathrm{baseline}}[i, j] + \varepsilon} \times 100\%. \tag{11}$$

The computed PDM illustrates the differences in Divergence Factors between the baseline and unknown states for selected bridges.

3. Calculate the Separation Factor (SF) for Bridge i to assess how distinctly the bridge separates from the others in terms of potential damage. The SF is computed as follows:

$$\mathrm{SF}_i = \frac{\sum_{k=1}^{m} \Delta \mathrm{DF}[i, k] + \sum_{k=1, k\neq i}^{m} \Delta \mathrm{DF}[k, i]}{\sum_{h=1}^{m} \sum_{k=1}^{m} \Delta \mathrm{DF}[h, k]}, \tag{12}$$

where SF_i represents the SF for Bridge i, $\Delta \mathrm{DF}[i, k]$ and $\Delta \mathrm{DF}[k, i]$ are the percentage differences for the corresponding bridge pairs, and m is the total number of bridges.

4. Next, compute the RDI for Bridge i in the unknown state using its calculated SF_i:

$$\mathrm{RDI}_i = e^{\mathrm{SF}_i} - 1. \tag{13}$$

5. Thresholds for each bridge are established using an internal cross-validation procedure applied to the baseline dataset. Specifically, the baseline dataset is randomly divided into two equal subsets multiple times. For each trial, one subset serves as the training set, while the other serves as a validation set. The SF and corresponding RDI values are computed for each validation set to generate a distribution of baseline RDI values.

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6. The threshold for Bridge i is statistically defined based on the mean and standard deviation (std) of these RDI values across all trials as follows:

$$\text{threshold}_i = \text{mean}\left(\text{RDI}_{i,\text{validation trials}}\right) + 2.5 \times \text{std}\left(\text{RDI}_{i,\text{validation trials}}\right). \quad (14)$$

7. Finally, compare the calculated RDI_i with the established threshold_i. If RDI_i is greater than threshold_i, Bridge i is considered damaged; otherwise, it is considered intact.

3. Computational Study

This study aims to develop a scalable SHM framework capable of accurately detecting structural anomalies across a network of bridges. The numerical modeling is designed to simulate dynamic interactions between a range of vehicle and bridge configurations, providing the basis for analyzing changes in bridge behavior. Specifically, these models capture vehicle-induced forces and bridge responses, which are integral to identifying potential structural damage and assessing bridge health within a multibridge inference system.

Each bridge is represented by a finite element (FE) model with varied properties to reflect real-world diversity in bridge design, including different lengths, cross-sectional areas, and moment of inertias. Vehicles are simulated as moving mass-spring-damper systems with distinct properties, including varying masses, speeds, stiffness levels and damping ratios. These simulations create a dynamic interaction between the vehicles and the bridges, producing the data necessary to detect structural changes across the network. By capturing these interactions, the model supports MISHM's goal of leveraging cross-bridge information to improve detection accuracy.

3.1. Modeling Techniques. To simulate the dynamic interaction between vehicles and bridges, this study employed a quarter car model as the primary vehicle representation. The quarter car model is widely used in VBI studies because it offers a simplified yet effective way to capture the essential vertical dynamic behavior of the vehicle without overly complicating the system. While more complex models, such as half-car or full-car models, can provide insights into lateral dynamics like roll and pitch or the geometric effects of multiple wheels, these factors are not the primary focus in this study. As noted in [44], the half-car and full-car models are often used when such lateral dynamics and geometric complexities are of interest. Therefore, the quarter car model was chosen for its balance between computational efficiency and accuracy, making it ideal for large-scale simulations in this research. The schematic diagram of the quarter car and bridge models used in this study is illustrated in Figure 5 [45], providing a visual overview of the primary components of the VBI framework.

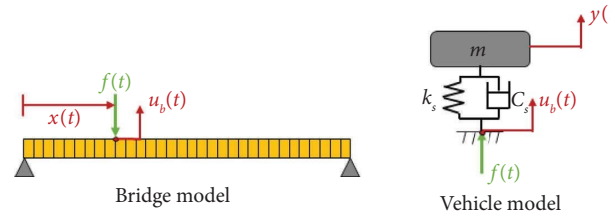


Figure 5: Schematic diagrams of the numerical VBI model.

The iterative VBI analysis follows a specific procedure, which is outlined in Figure 6. At each time step t, the bridge's vertical response is evaluated under the vehicle's previous contact force $f(t-1)$, which is now applied at the new wheel location $x(t)$. The initial conditions for the bridge response are set as $w_b(t-1)$. After the bridge analysis is completed, the updated bridge response vector $w_b(t)$ provides an estimate of the bridge displacement at the wheel's location $u_b(t)$. This displacement is used as base excitation for the single-degree-of-freedom (SDOF) OpenSees vehicle model with a fixed base to obtain the vehicle's vertical response $y(t)$. The updated vehicle response determines the new contact force $f(t)$, and the iterative process continues until convergence is achieved. The procedure then advances to the next time step.

3.2. Properties of Bridges and Vehicles. For the bridge models, structures were simulated with variations in cross-sectional area, moment of inertia, and length to reflect typical bridge characteristics encountered in practice. All bridges share consistent values for Young's modulus, discretization points, and material density, chosen to represent common values for reinforced concrete structures [22, 46–49]. Detailed bridge parameters and corresponding first three natural frequencies are presented in Tables 1 and 2.

Road roughness significantly impacts bridge responses and is recognized as one of the primary challenges in indirect mode shape identification [50]. To address this, a low-level road roughness profile was generated for each bridge using the power spectral density (PSD) method described in [51], thus adding imperfections to the bridge surface. An example road roughness profile generated for Bridge 1 is illustrated in Figure 7.

Vehicles simulated in this study varied in mass, stiffness, speed, and damping based on truncated normal statistical distributions to reflect typical traffic conditions encountered on urban bridges. For each bridge and each structural state (intact and damaged), simulations were conducted using 1000 unique vehicles. These distributions, detailed in Table 3, are aligned with commonly adopted parameters in the VBI literature [22, 29, 46–49, 52]. Utilizing such extensive simulations facilitates a comprehensive analysis of the dynamic impacts that diverse vehicle configurations have on bridge responses. Histograms illustrating the parameter distributions are provided in Figure 8.

A representative example of the simulated vehicle parameter distributions for Bridge-1 is illustrated in Figure 8.

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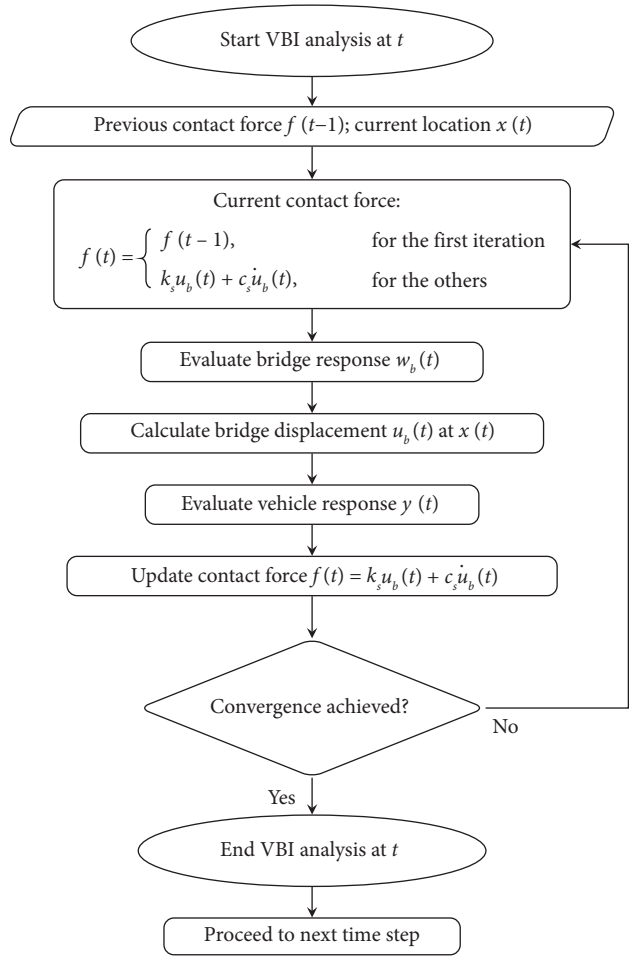


Figure 6: Flowchart of the iterative VBI analysis for time t.

3.3. Data Extraction and Preprocessing Phase. The proposed MISHM model assumes that the statistical distributions of various vehicle types passing over all bridges in the system are identical. To reflect diverse traffic conditions, 1000 unique vehicles were randomly generated following the truncated normal distributions described in Section 3.2 for each bridge and each structural state (baseline and damaged scenarios). Furthermore, to simulate background traffic-induced vibrations, each simulation run additionally included 15 to 25 randomly generated vehicles following the same statistical distributions.

During these simulations, vertical acceleration signals $x(t)$ at the vehicle locations on each bridge were extracted at a sampling frequency of 100 Hz, which is typical in VBI studies for capturing dynamic response signals [24, 27, 32, 34, 35, 53]. The acceleration signals were assumed to be recorded by either mobile devices (e.g., smartphones securely mounted on vehicle dashboards) or vehicle-integrated accelerometer sensors. In both cases, sensors record vertical accelerations consistent with vehicle-body movements, capturing responses influenced by VBI dynamics. By comfortably exceeding the Nyquist sampling criterion, a sampling frequency of 100 Hz ensures accurate capture of dominant dynamic characteristics and damage-related features of the bridges without aliasing. To replicate real-world measurement conditions, 10% artificial noise was added to all signals to mimic environmental disturbances such as wind effects and sensor inaccuracies. Consequently, comprehensive datasets consisting of 1000 signals per bridge per structural state were generated for subsequent analysis. Rather than explicitly extracting bridge natural frequencies from these acceleration signals, MFCCs derived from the recorded accelerations were analyzed statistically using KL divergence, as detailed in the Methodology section, facilitating indirect yet effective extraction of bridge-specific dynamic features from combined vehicle-bridge responses.

During the monitoring phase, the same data collection and preprocessing procedures were repeated. Two distinct damaged states were explicitly considered and simulated on Bridge-5: a minor damage scenario (hereafter referred to as Damage Case 1) involving a localized stiffness reduction of 20%, and a more severe damage scenario (Damage Case 2) involving a localized stiffness reduction of 40%. Both damage scenarios affected a segment centered at the mid-span, covering 10% of the bridge's total length. For threshold establishment and validation, an internal cross-validation

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Table 1: Baseline properties of simulated bridges.

Parameter	Value	Notes
Young's modulus E_b	27.5×10^9 Pa	Young's modulus
Density ρ	2400 kg/m^3	Typical concrete density
Damping ratio ζ	1.5%	Damping fraction
Number of discretization points n	500	Finite-element discretization points

Table 2: Simulated bridge parameters and corresponding natural frequencies.

Bridge ID	Length (m)	Area (m^2)	Moment of inertia (m^4)	f_1 (Hz)	f_2 (Hz)	f_3 (Hz)
Bridge-1	25.0	2.00	0.10	1.902	7.609	17.121
Bridge-2	25.0	2.00	0.12	2.084	8.336	18.755
Bridge-3	25.0	2.25	0.10	1.794	7.174	16.142
Bridge-4	25.0	2.25	0.12	1.965	7.859	17.682
Bridge-5	27.5	2.00	0.10	1.572	6.289	14.150
Bridge-6	27.5	2.00	0.12	1.722	6.889	15.500
Bridge-7	27.5	2.25	0.10	1.482	5.929	13.340
Bridge-8	27.5	2.25	0.12	1.624	6.495	14.614
Bridge-9	30.0	2.00	0.10	1.321	5.284	11.890
Bridge-10	30.0	2.00	0.12	1.447	5.789	13.024
Bridge-11	30.0	2.25	0.10	1.246	4.982	11.210
Bridge-12	30.0	2.25	0.12	1.364	5.458	12.279

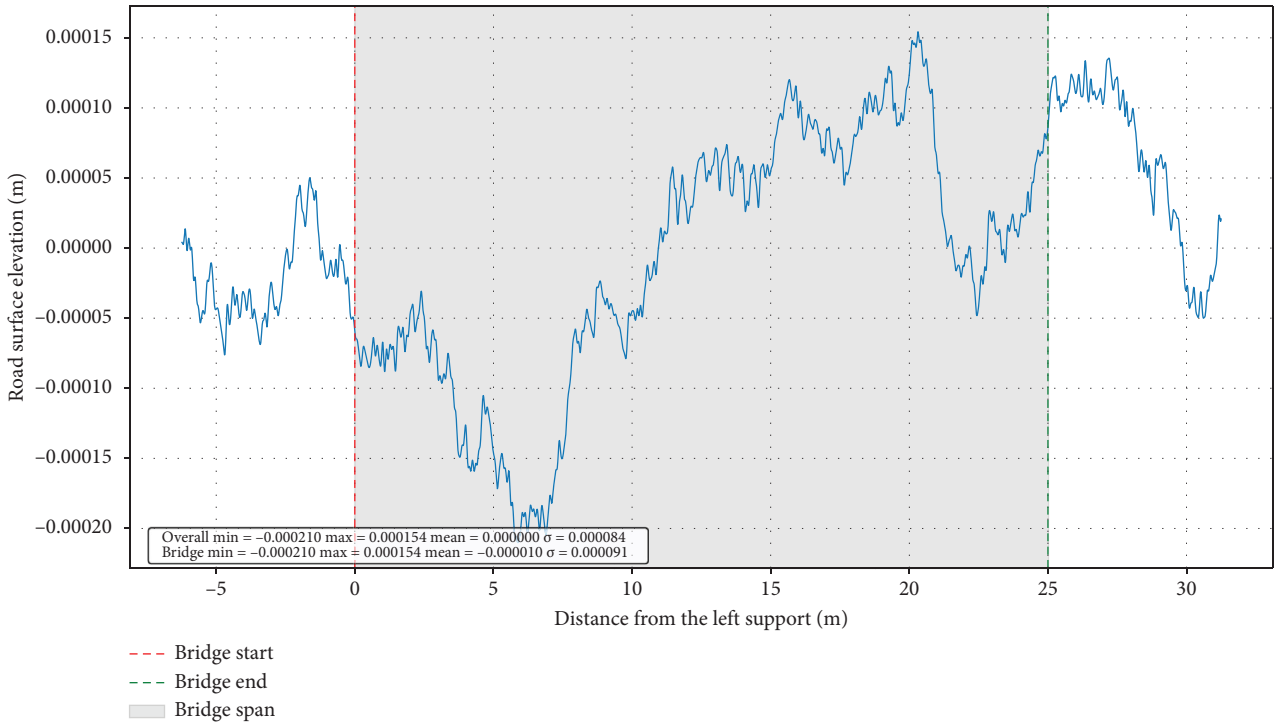


Figure 7: Example of road roughness profile generated for Bridge-1.

Table 3: Probability distributions of simulated vehicle parameters.

Parameter	Probability distribution	Min	Max	Mean	Standard deviation	Unit
Mass	Truncated normal	800	1200	1000	50	kg
Stiffness	Truncated normal	80,000	100,000	90,000	1000	N/m
Speed	Truncated normal	10	15	12.5	1.25	m/s
Damping	Truncated normal	4000	6000	5000	250	N· m/s

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procedure was applied directly to the baseline (intact) dataset. Specifically, the baseline dataset of 1000 samples was randomly split into two equal subsets multiple times. In each trial, one subset was designated as the training set (baseline), and the other subset as the validation set.

4. Results and Discussion

4.1. PDM. The PDMs serve as essential tools in bridge health monitoring, offering insights into variability between the validation and unknown states. MISHM utilizes KL Divergence to generate nonsymmetrical matrices, reflecting the directional nature of divergence between bridges i and j. Despite this inherent asymmetry, both $\Delta \mathrm{DF}[i, j]$ and $\Delta \mathrm{DF}[j, i]$ are considered when calculating the RDI for each bridge, ensuring accurate assessment.

For this analysis, two damage cases were examined on Bridge-5 to illustrate MISHM's sensitivity across different damage severities: Damage Case 1 (20% stiffness reduction) and Damage Case 2 (40% stiffness reduction). The validation PDM shown in Figure 9 displays minimal variability, confirming stable conditions.

In the two damage cases, Bridge-5 exhibits clear, progressive increases in percentage differences, as illustrated in Figures 10 and 11. With higher severity (Damage Case 2), the PDM entries related to Bridge-5 become significantly more pronounced. This progression demonstrates that MISHM effectively captures both minor and severe structural anomalies.

Together, these insights indicate that MISHM not only detects damage sensitively across a range of severities but also gains reliability as the number of bridges increases, making it especially well-suited for large-scale structural-health–monitoring campaigns and proactive bridge-maintenance strategies.

4.2. Statistical Summaries of PDM Differences. To further examine the differences across the baseline and damaged states, statistical summaries of the PDM entries for Bridge-5 under two different damage severity levels were generated (Figure 12). This box plot presents a statistical summary of the PDM values, offering insights into the distribution and variability of differences across these states. In the box plot, the central line represents the median value, indicating central tendency. The box itself encapsulates the interquartile range (IQR), representing the middle 50% of the data, while the whiskers extend to the minimum and maximum values within a specified range, illustrating the overall spread of the data.

The validation state exhibits tightly clustered values near 2.0%, indicating stable and intact conditions. Damage Case 1 shows an increased distribution of differences with a median around 8.0%, clearly distinguishing it from the baseline state. Damage Case 2 demonstrates an even more dispersed distribution with a significantly higher median, around 17.5%, highlighting a clear progression in damage severity.

These findings validate MISHM's capability to detect and differentiate varying levels of structural damage severity, affirming its effectiveness as a tool for SHM and proactive bridge maintenance strategies.

4.3. RDI and Threshold Results. The RDI and its corresponding threshold quantify each bridge's deviation from baseline behavior. To illustrate the method's sensitivity across our two damage scenarios on Bridge-5, we plot the RDI ratios (normalized by their thresholds) for both Damage Case 1 and Damage Case 2 in Figure 13. Each marker represents one RDI calculation from a separate cross-validation trial.

As shown, all RDI-to-threshold ratios exceed 1.0 for both cases, confirming that the framework reliably flags Bridge-5 as damaged under both minor (Damage Case 1) and severe (Damage Case 2) conditions. Furthermore, Damage Case 2 produces systematically higher RDI ratios than Damage Case 1, demonstrating the method's ability not only to detect damage but also to reflect its severity quantitatively.

4.4. Influence of the Number of Vehicles. A critical factor influencing the practical application of the MISHM framework is the number of vehicles utilized during monitoring. To assess how vehicle sample size affects the consistency of damage detection, the coefficient of variation of RDIs was analyzed across varying numbers of vehicles. For each selected sample size, 30 independent trials were performed to ensure statistical robustness.

Figure 14 illustrates the relationship between vehicle sample size and the corresponding coefficient of variation of the RDIs for Bridge-5 under Damage Case 1 and Damage Case 2. A clear trend emerges, indicating that an increase in the number of vehicles notably decreases the coefficient of variation of RDIs, thus enhancing the reliability of damage detection outcomes. Additionally, the consistently lower variation observed for Damage Case 2 (across all sample sizes except the maximum sample size) demonstrates that more severe damage is easier to detect reliably.

Therefore, it is safe to conclude that the proposed MISHM approach demonstrates significantly improved robustness with increasing numbers of vehicles. This finding is particularly promising for real-life applications, where big data can be extracted from extensive fleets of vehicles, thus facilitating consistent and accurate SHM outcomes.

It is important to note that single-bridge monitoring approaches, which compare a single structure only to its own baseline, can still detect damage under certain conditions. By contrast, MISHM's network-level perspective is inherently aligned with large-scale urban infrastructure monitoring, accommodating multiple structures and the variability that comes with broader system-wide assessments. This broader viewpoint supports reliable evaluation across extensive bridge networks, without necessarily negating the value of single-bridge approaches in more localized contexts.

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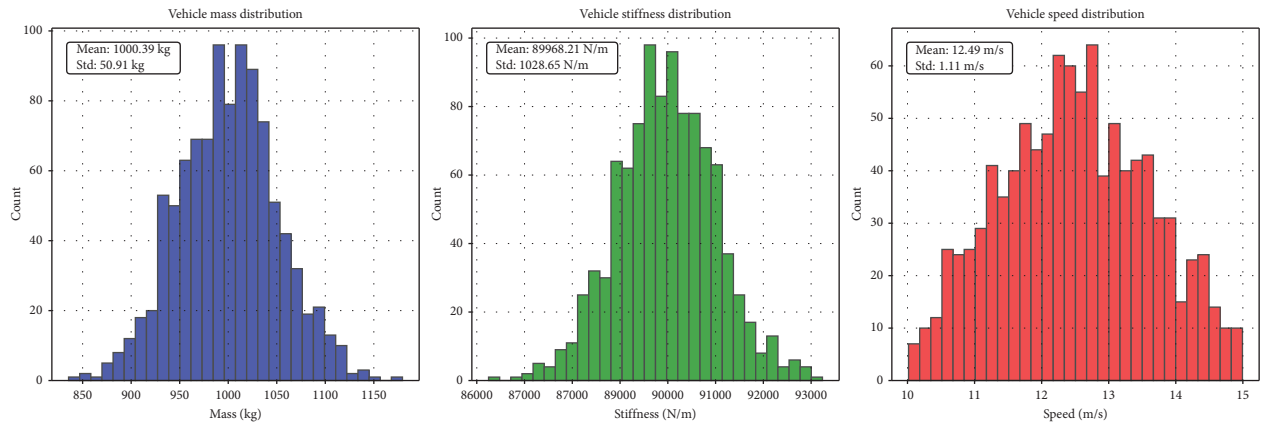


FIGURE 8: Histograms illustrating the distributions of vehicle mass, stiffness, speed, and damping for simulations performed on Bridge-1 in the intact state.

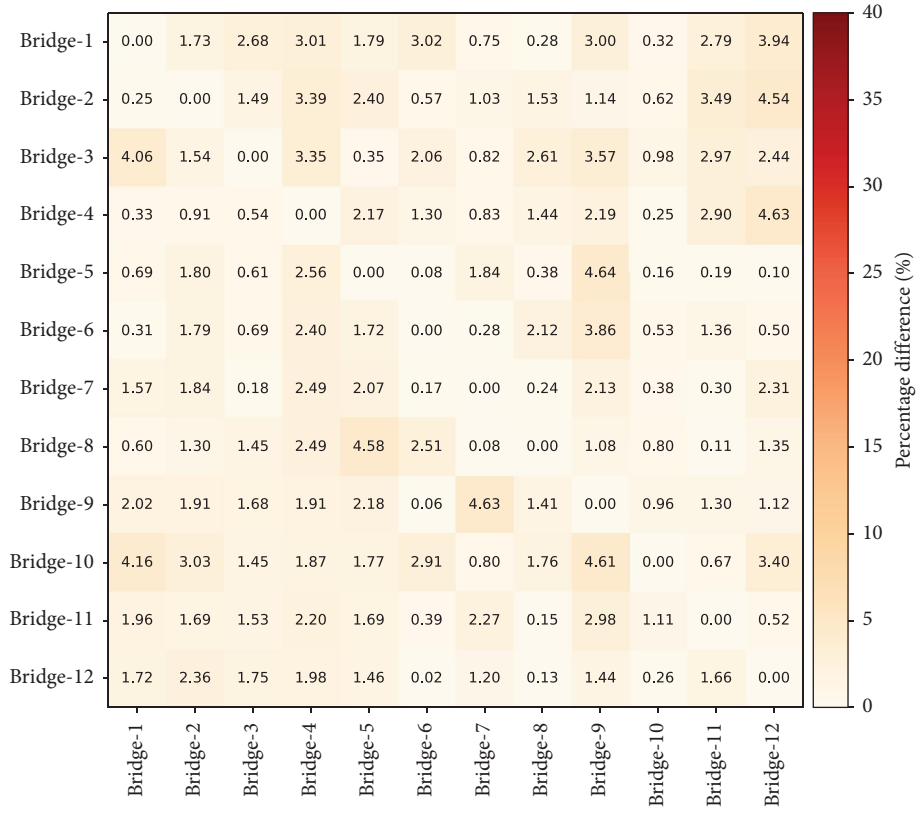


FIGURE 9: PDM representing all bridges in the validation state.

4.5. Sensitivity to Small-Magnitude Damage. To delineate the lower bound of detectability of the proposed framework, two small-magnitude damage scenarios were introduced on Bridge-5:

- Damage Case Small 1: 10% stiffness loss over 5% span length at the 1/4 span, a location inherently less impactful on the global dynamic response, thus more challenging to detect.

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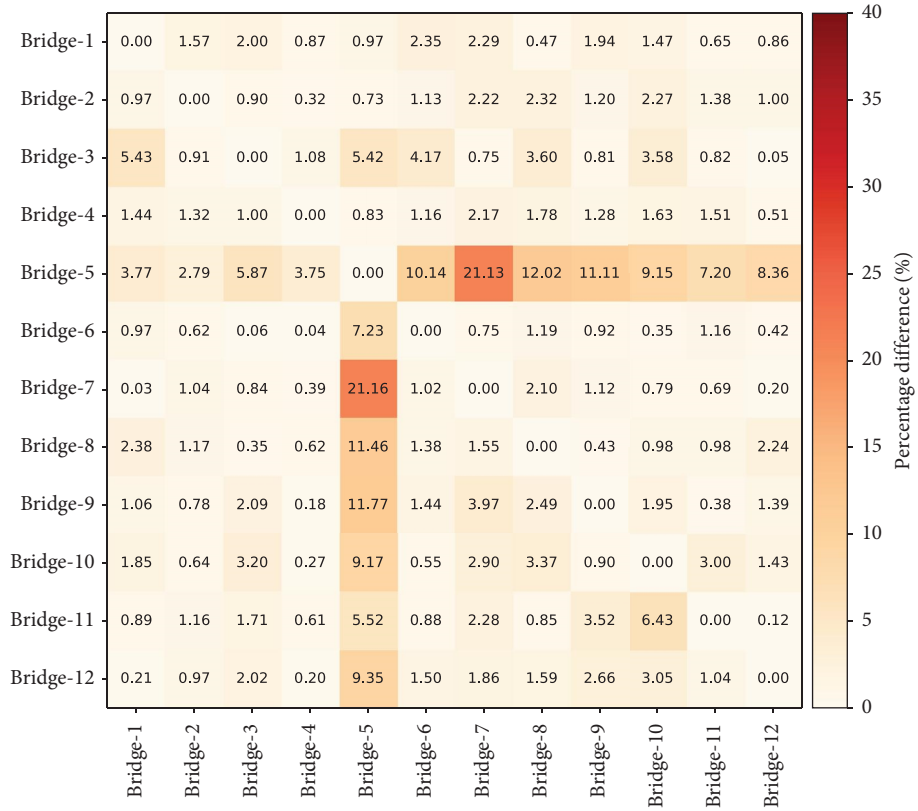


Figure 10: PDM for Damage Case 1 on Bridge-5.

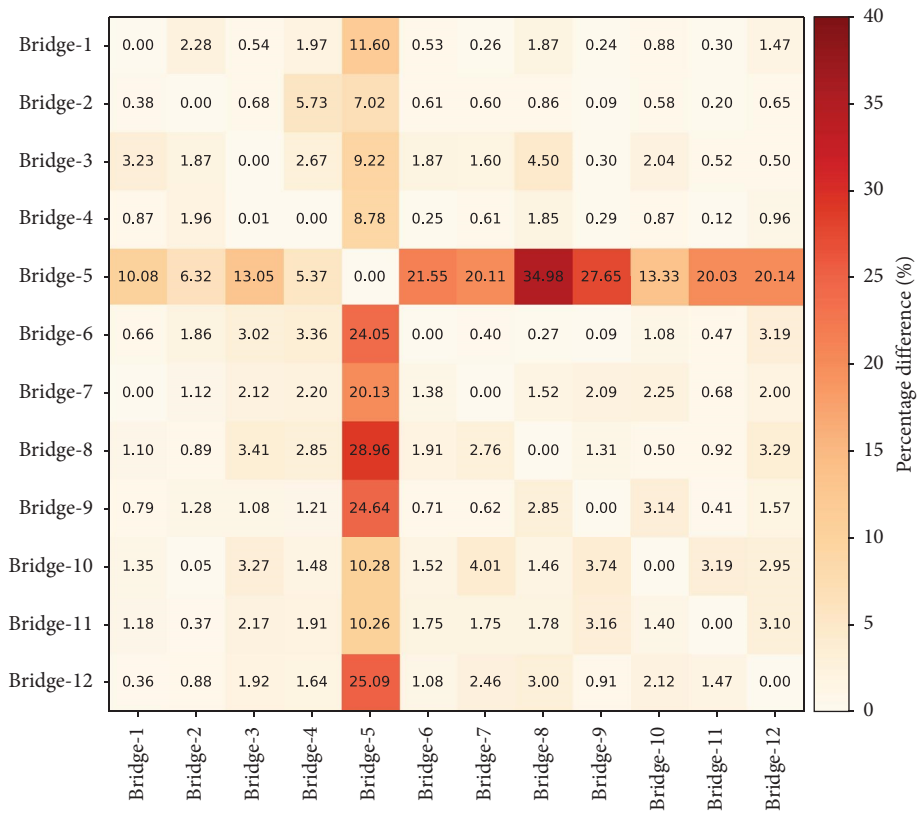


Figure 11: PDM for Damage Case 2 on Bridge-5.

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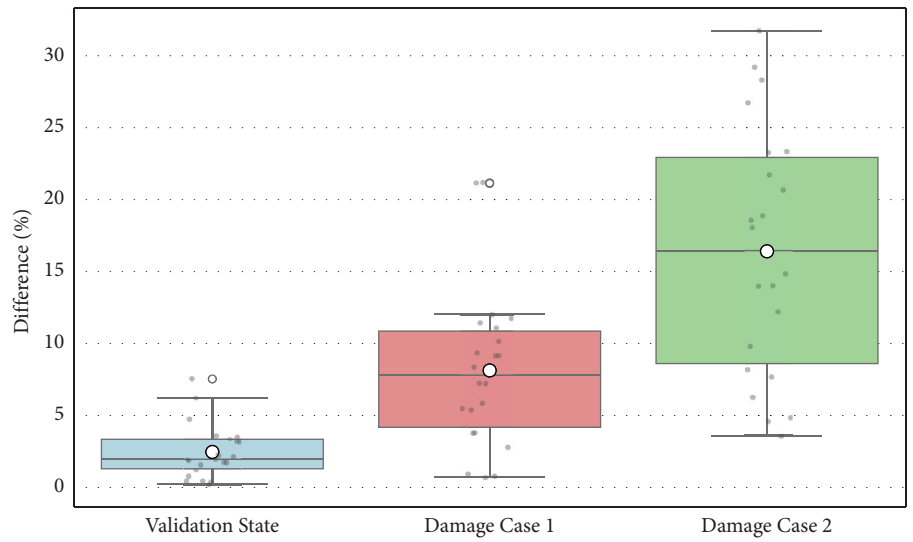


FIGURE 12: Comparative statistical summaries of PDM differences for Bridge-5 across validation state, Damage Case 1, and Damage Case 2.

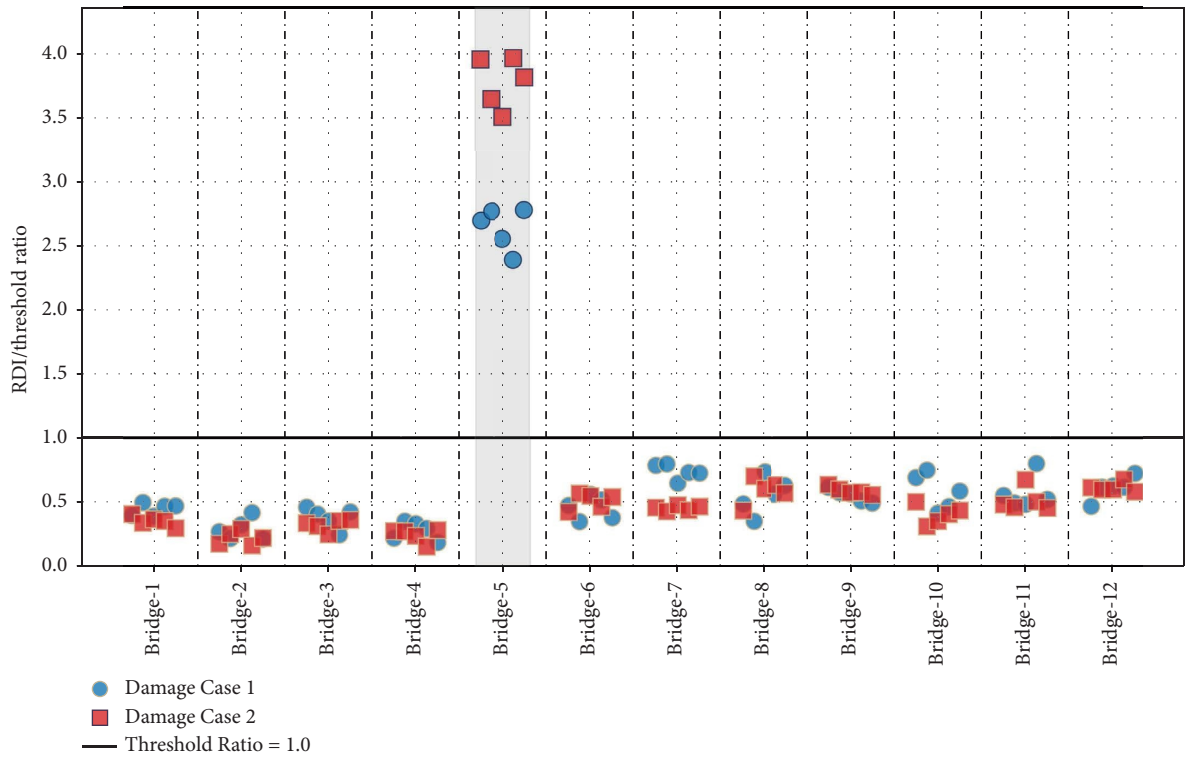


FIGURE 13: Bridge-5 RDI-to-threshold ratios for Damage Case 1 (20% stiffness loss) and Damage Case 2 (40% stiffness loss). Markers above the horizontal line at 1.0 indicate detected damage.

- Damage Case Small 2: a multiple-damage scenario consisting of two separate 10% stiffness losses—each over 5% of the span—located at 1/4 span and 1/2 span.

The corresponding percentage differences compared to the validation baseline are presented as box plots in Figure 15, followed by the RDIs for these two cases illustrated in Figure 16.

The box plots demonstrate a slight increase in percentage differences for the multiple-damage scenario (Damage Case Small 2) compared to the single-damage scenario (Damage Case Small 1). However, this increase is relatively limited, indicating that multiple minor stiffness reductions do not substantially amplify the global dynamic response.

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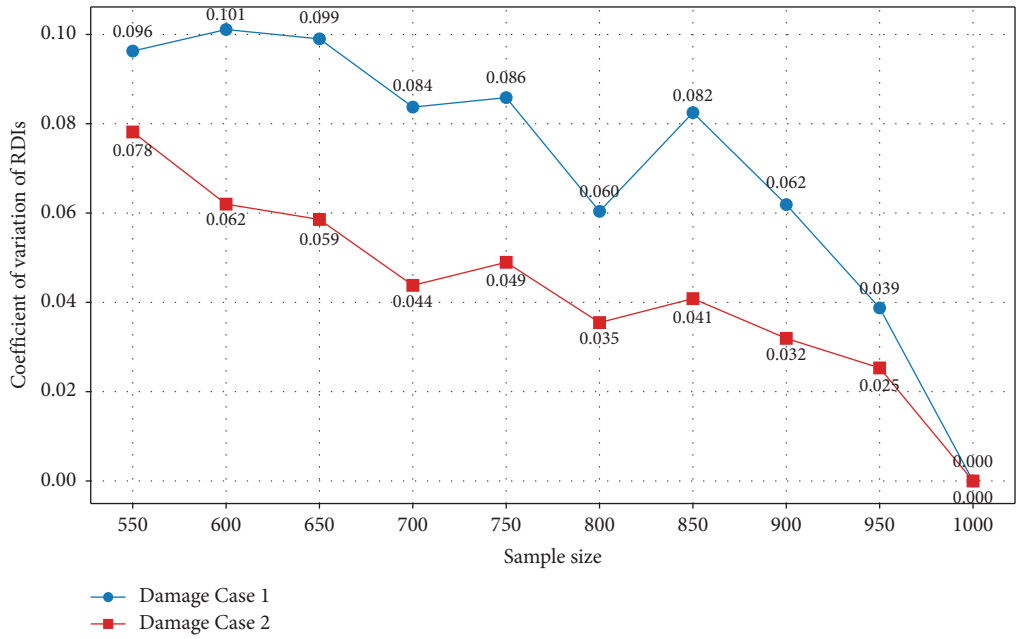


FIGURE 14: Coefficient of variation of RDIs versus number of vehicles for Bridge-5 under Damage Case 1 and Damage Case 2.

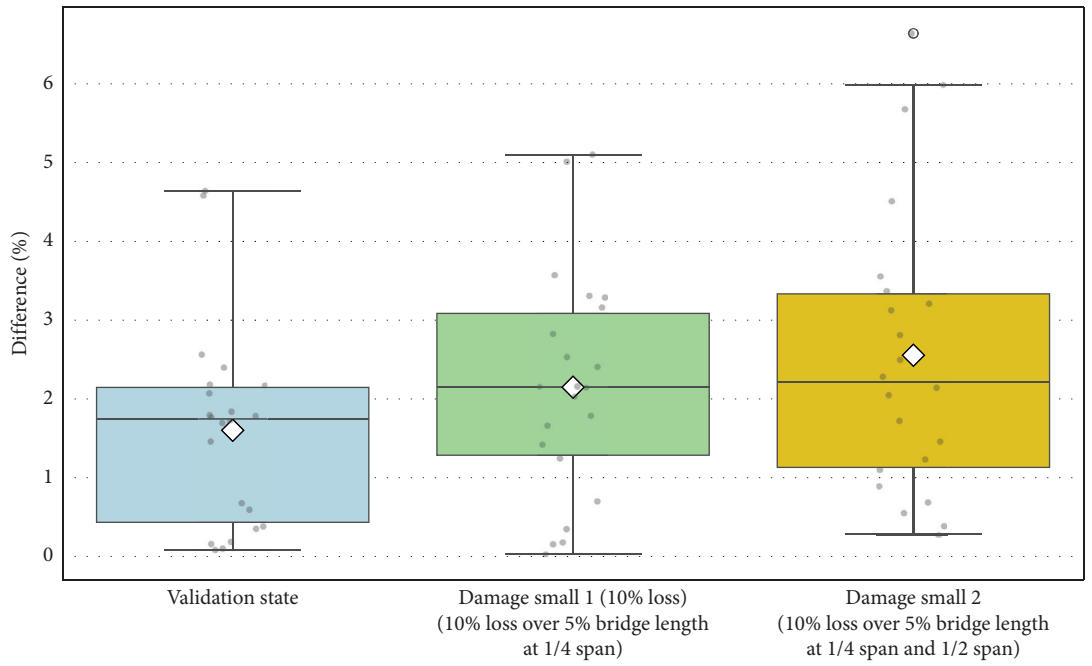


FIGURE 15: Boxplot comparison of percentage differences from validation baseline for the two small-magnitude damage scenarios.

Subsequently, the RDIs for both scenarios are marginally above the detection threshold, as seen in Figure 16. This confirms the sensitivity of the proposed approach to subtle stiffness reductions. The similarity between the RDI results for both scenarios indicates that the method effectively detects minor structural changes by identifying the presence

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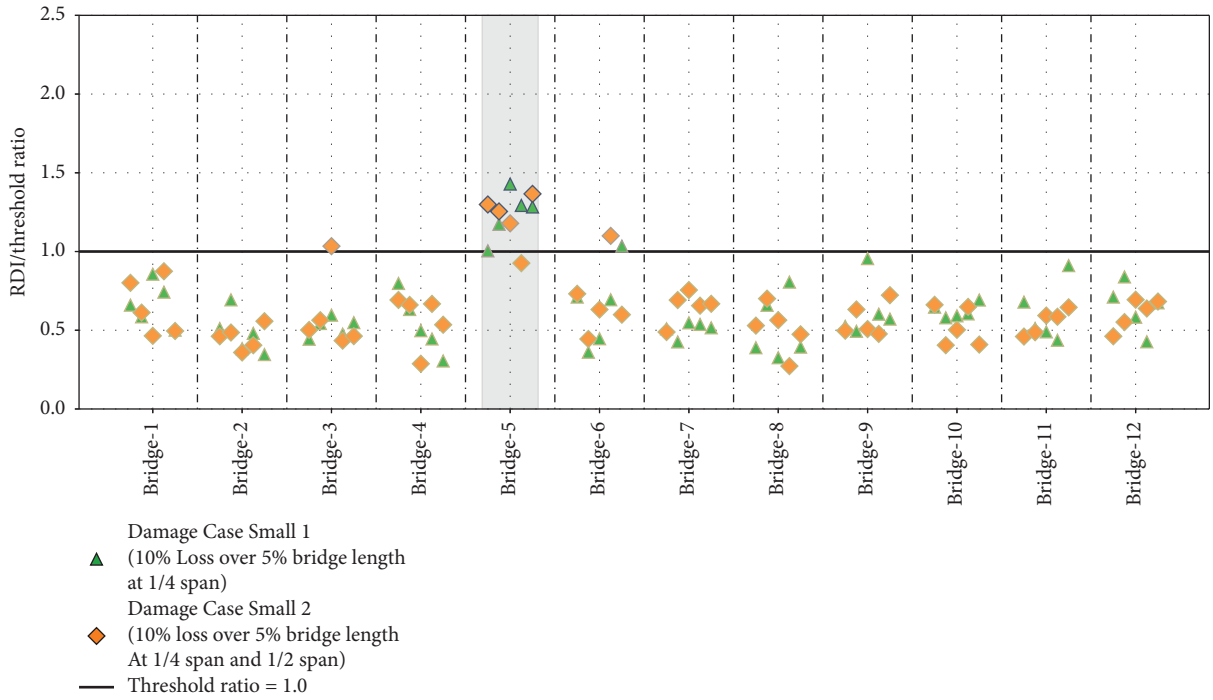


Figure 16: RDI-to-threshold ratios for the two small-magnitude damage scenarios on Bridge-5.

of damage within a bridge network context, aligning well with the intended scope of the current framework, which emphasizes network-level detection rather than precise damage localization.

5. Conclusions and Future Works

This study introduces the MISHM framework, integrating drive-by crowdsensing data with advanced analysis techniques using MFCCs and KL Divergence. Comprehensive numerical studies were conducted, focusing specifically on Bridge-5 under VBI conditions with random traffic scenarios. Two distinct damage scenarios were simulated: Damage Case 1, representing a minor stiffness reduction (20%), and Damage Case 2, signifying a severe stiffness reduction (40%). Both damage cases affected a segment covering 10% of the total bridge span at mid-span.

The results demonstrated that Damage Case 1 yielded a median percentage divergence difference of approximately 8%, while Damage Case 2 exhibited a significantly higher median difference of around 17.5%. These clear differences underscore the sensitivity and precision of MISHM in effectively distinguishing between different levels of structural damage severity. Furthermore, the RDI values for both damage cases consistently surpassed their predefined thresholds, reinforcing MISHM's capability for reliable damage detection.

PDMs provided intuitive visual diagnostics, highlighting structural health variations clearly across the monitored bridge network. Such visual assessments enhance the interpretability of structural health outcomes, enabling rapid identification and prioritization of maintenance interventions.

Additionally, the influence of vehicle sample size on detection performance was rigorously evaluated. Results indicated that the dispersion of RDIs, measured by the coefficient of variation, decreased significantly with increased sample sizes. Consequently, it is safe to conclude that MISHM performs robustly when leveraging larger datasets from extensive vehicle fleets, aligning effectively with practical applications involving big data from crowdsensing platforms.

The sensitivity analysis further demonstrated the framework's capability to detect subtle structural changes, as highlighted by two additional small-magnitude damage scenarios: Damage Case Small 1, involving a single stiffness reduction (10%) across 5% of the span length at the 1/4 span of Bridge-5, and Damage Case Small 2, a multiple-damage scenario with two separate stiffness reductions (each 10%) at the 1/4 span and 1/2 span of Bridge-5. The RDIs for these scenarios were marginally above the detection threshold, confirming the sensitivity and reliability of MISHM in identifying minor structural deterioration. These findings emphasize the framework's potential as an effective early-warning tool within a bridge network context, complementing existing monitoring strategies.

Despite these promising findings, certain limitations remain. First, environmental factors (e.g., temperature fluctuations and wind effects) were not included in the current simulations, potentially introducing measurement uncertainties or false positives in practical scenarios. Second, the current methodology primarily considers limited multiple-damage scenarios, highlighting the necessity to further investigate more complex and simultaneous damage patterns.

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Future research directions will focus initially on experimental validation through controlled laboratory tests and subsequent field investigations explicitly considering environmental variations. Additionally, extending the MISHM framework to effectively handle multiple damaged bridges and diverse damage patterns within bridge networks will be pursued to further enhance its scalability and applicability in large-scale SHM.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Conflicts of Interest

The authors declare no conflicts of interest.

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